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Guideline/Protocol Title	**Guidelines for the Management of Urinary Tract Infections in Children**
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Committee Review	Antimicrobial Stewardship Subcommittee Pharmacy and Therapeutics Committee
Target Population	Pediatric patients with suspected urinary tract infections
Overview	This guideline provides evidence-based management of suspected urinary tract infections.
Effective Date	12/02/2021
Revised Date	1/2024; 11/2024
Expiration Date	11/2027
Schedule for Periodic Review	Every 3 years
Implementation Strategy	See below. Each service will be trained on the guideline.
Education Strategy	ID pharmacist, ID physician, and ASC service line representatives will educate each affected service. Email communication will be sent as well.
Primary Outcome(s)	Selection of antimicrobial therapy and duration of therapy.
Outcome Assessment Plan	
Document Posting Location	☒ CareWeb Guidelines and Protocols ☐ Markey Cancer Center (MCC) Protocols ☒ Other Location: Antimicrobial Stewardship site
Information Technology Needs	☒ **Link to LexiDrugs** – *Medication(s): nitrofurantoin, cefazolin, cefadroxil, cephalexin, ceftriaxone, cefepime, ertapenem, TMP/SMX, ciprofloxacin, amoxicillin, amoxicillin/clavulanate, ampicillin, linezolid, levofloxacin*

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Purpose of Guidelines: To provide evidence-based guidelines for the appropriate management of pediatric patients aged >60 days with urinary tract infections (UTIs) presenting to the Emergency Department or admitted to UK HealthCare (UKHC).

Background: Multiple guidelines have been published for the diagnosis, prevention, and treatment of urinary tract infections, one of the most common indications for antimicrobials. The American Academy of Pediatrics (AAP) created a clinical practice guideline in 2011 and reaffirmed the document in 2016 to describe the diagnosis and management of an initial UTI in febrile infants and children aged 2-24 months.

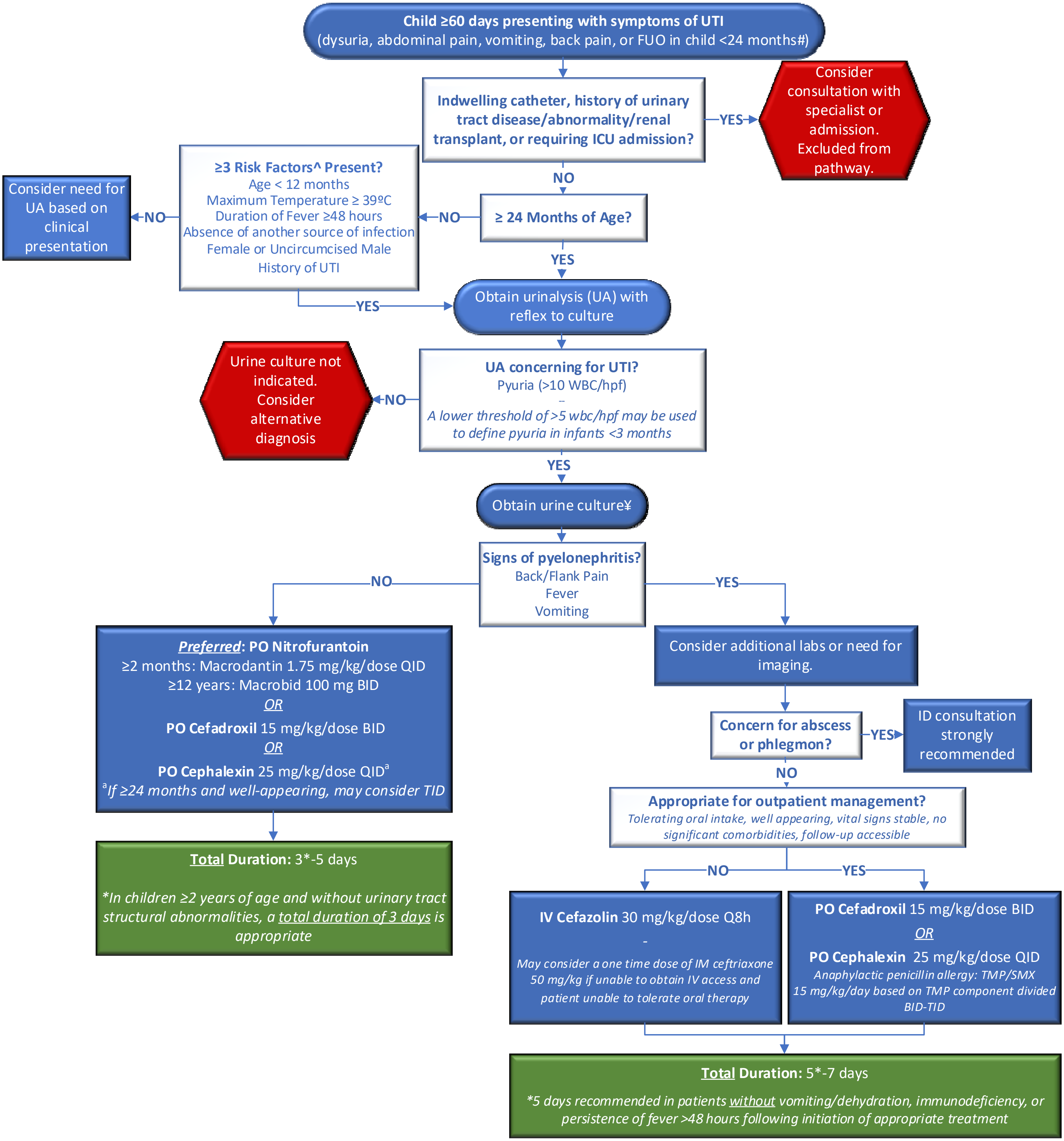
Diagnosis:

- Obtain urinalysis (UA) only in pediatric patients with symptoms of UTI (i.e., urinary urgency, dysuria, suprapubic pain, fevers with emesis, fevers with unknown origin in a child <24 months).
    - "Foul smelling" or "cloudy" urine has poor correlation with a UTI and should <u>NOT</u> be used as criterion to obtain a UA or urine culture.
- Urine cultures:
    - Toilet-trained patients: mid-stream clean catch urine cultures (after preparation)
    - Non-toilet-trained patients: catheterized urine cultures
    - The culturing of bagged specimens or indwelling urinary catheters is discouraged as it may be representative of normal GU, GI, and skin colonization.

Antimicrobial stewardship principles associated with urinary tract infection management:

- To establish the diagnosis of UTI, the presence of <u>BOTH</u> a positive urinalysis (demonstrating pyuria and/or bacteriuria) <u>AND</u> the growth of ≥50,000 CFU/mL of a uropathogen from urine culture AND symptoms consistent with UTI are required.
- **Antibiotics should <u>NOT</u> be initiated for treatment of UTIs in the absence of symptoms.**
    - The only two populations with data supporting treatment of asymptomatic bacteriuria (ASB) include (1) pregnancy, and (2) preparation for a planned urinary tract procedure.
    - Symptoms may be challenging to interpret in children <24 months of age (particularly in those who are preverbal). See Figure 1 for risk stratification to guide evaluation.
- Always review previous urine cultures to assist with selection of empiric therapy.
- Always narrow antibiotics when possible and stop therapy if urine cultures demonstrate no growth.
- Prompt de-escalation from intravenous (IV) to oral/enteral is appropriate in patients able to tolerate PO and demonstrating clinical improvement.
- Per CLSI laboratory standards, <u>**cefazolin**</u> serves as the surrogate to determine susceptibility for all oral cephalosporins for treatment of *E. coli, K. pneumoniae,* and *P. mirabilis* isolated from the urine.
    - Rates of susceptibility are identical for oral third and first generation cephalosporins.
    - Third generation cephalosporin use contributes significantly to the development of Gram-negative resistance, including ESBL-production.
- Do not send repeat urinalyses or urine cultures to evaluate for "test of cure".
- ID consultation is strongly recommended for treatment of any patient with renal abscess.

Figure 1: Pediatric Emergency Department and Inpatient Diagnosing and Empiric Treatment Algorithm



Consider sexual history and STI testing in patients ≥12 years of age; **#:** "Foul smelling" or "cloudy" urine has poor correlation with UTI and should NOT be used as criterion to obtain UA or urine culture; **^:** For a more detailed risk stratification, consider the University of Pittsburgh UTICalc; **¥:**Catheter specimens recommended for non-toilet-trained children; Urine culture with growth of **≥50,000 cfu/mL of a single uropathogen** considered consistent with UTI (urine cultures not meeting this criteria unlikely to be clinically significant and provider should consider colonization versus contamination). **∞:** 2-4 days of therapy have demonstrated noninferiority when compared to 7-14 days of treatment for cystitis in patients aged 3 months-18 years.

Table 1: Empiric Treatment for Pediatric Patients with Suspected UTI

CATEGORY	DEFINITION	EMPIRIC THERAPY
Asymptomatic Bacteriuria	No symptoms of a UTI even if pyuria and positive urine culture *Obtaining urine cultures in asymptomatic patients, regardless of presence of a catheter is **NOT** recommended*	NONE • If a patient is about to undergo a urologic procedure or is pregnant, treat like cystitis. • If a patient has an indwelling catheter, discontinue or change catheter if possible.
Cystitis	Dysuria, urgency, frequency, or suprapubic pain in the absence of fever or other systemic symptoms **AND** Pyuria* (>10 WBC/hpf) **AND** Positive urine culture (≥50,000 cfu/mL of a single uropathogen) **AND** Blood cultures (if obtained) negative for the organism isolated from urine	**PO Nitrofurantoin**[a] (preferred) *OR* **PO Cefadroxil** *OR* **PO Cephalexin** **Duration**: 3*-5 days **In patients ≥2 years of age and without urinary tract structural abnormalities, a total duration of 3 days is appropriate*

Pyelonephritis	Fever, flank pain, or ill appearance
	AND
	Pyuria* (>10 WBC/hpf)
	AND
	Positive urine culture (≥50,000 cfu/mL of a single uropathogen)
		IV Cefazolin (preferred)
		• **If able to tolerate PO:**
		  ○ PO Cephalexin
		  ○ PO Cefadroxil
		Alternative if no IV access: IM Ceftriaxone
		Although IV cefazolin is the preferred parenteral agent, IV ceftriaxone may be considered in patients being discharged from the ED if unable to re-dose antibiotics within 6 hours of discharge.
		Alternative if history of *Pseudomonas*: IV Cefepime
		Alternative if history of ESBL: IV Ertapenem
		• **If ESBL and susceptibilities known:**
		  ○ **Preferred:** PO TMP/SMX
		  ○ **If TMP/SMX resistant:** PO ciprofloxacin
		Duration: 5-7 days

*****: a lower cutoff of >5 wbc/hpf should be used to define pyuria in infants <3 months; **a**: Nitrofurantoin provides excellent empiric coverage and concentrates well in the bladder but should not be used in children <2 months of age or any child with suspicion of urosepsis or pyelonephritis.

Table 2: Follow up for Pediatric Patients with Suspected UTI

RECOMMENDED FOLLOW UP:
• If non-pretreated urine culture is negative, discontinue antibiotics. • Growth of ≥1 organism in a non-catheterized patient is unlikely to be clinically significant. Consider discontinuing antibiotics in this scenario. • Adjust empiric therapy according to susceptibilities to narrow targeted therapy. • Febrile infants (≤24 months) presenting with first UTI should undergo a screening renal and bladder ultrasonography (RBUS). RBUS should be obtained within the first 2 days of treatment initiation in infants with unusually severe illness or when substantial clinical improvement is not noted. Infants who demonstrate substantial clinical improvement should have RBUS obtained shortly following the completion of therapy for the acute episode. • Febrile infants presenting with recurrent UTI may warrant additional imaging and/or follow up with urology.

Table 3: Dosing of and Activity of Definitive Treatment Options

ANTIMICROBIAL	DOSING IN PATIENTS ≥1 MONTH OF AGE WITH NORMAL RENAL FUNCTION	% SUSCEPTIBLE PER KCH 2023-2024 ANTIBIOGRAM	
		E. coli	*P. aeruginosa*
Amoxicillin	45 mg/kg/DAY PO divided Q8h (max dose: 500 mg)	44%	R
Amoxicillin/ Clavulanate	45 mg/kg/DAY PO divided Q8h (max dose: 500 mg)	51%	R
Ampicillin	50 mg/kg/dose IV Q6h (max dose: 2000 mg)	44%	R
Cefadroxil	15 mg/kg/dose PO Q12h (max dose: 1000 mg)	94%	R
Cefazolin	30 mg/kg/dose IV Q8h (max dose: 2000 mg)	94%	R
Cefepime	50 mg/kg/dose IV Q8h (max dose: 2000 mg)	96%	94%
Ceftriaxone	50 mg/kg/dose IV/IM Q24h (max dose: 2000 mg)	93%	R
Cephalexin	Cystitis: 25 mg/kg/dose PO Q6h (max dose: 500 mg) Pyelonephritis: 25 mg/kg/dose PO Q6h (max dose: 1000 mg)	94%	R
Ciprofloxacin	20 mg/kg/dose PO Q12h (max dose: 750 mg)	74%	100%
Ertapenem	<12 years: 15 mg/kg/dose IV Q12h (max dose: 500 mg) ≥12 years: 15 mg/kg/dose IV Q24h (max dose: 1000 mg)	100%	R
Linezolid	<12 years: 10 mg/kg/dose IV/PO Q8h (max dose: 600 mg) ≥12 years: 10 mg/kg/dose IV/PO Q12h (max dose: 600 mg)	-	-
Levofloxacin	≤5 years: 10 mg/kg/dose PO Q12h (max dose: 500 mg) >5 years: 10 mg/kg/dose PO Q24h (max dose: 750 mg)	78%	94%
Nitrofurantoin[a] (NOT for pyelonephritis)	≥1 month: Macrodantin 1.75 mg/kg/dose PO Q6h (max dose: 100 mg) ≥12 years: Macrobid 100 mg PO Q12h	94%	-
Trimethoprim/ Sulfamethoxazole[b] (TMP/SMX)	Cystitis: 6 mg/kg/dose PO Q12h (max dose: 160 mg) Pyelonephritis: 15 mg/kg/DAY PO divided Q8-12h (max dose: 320 mg)	67%	R

[-]: drug not tested nor indicated; **R:** intrinsic resistance

[a] Nitrofurantoin provides excellent empiric coverage and concentrates well in the bladder, but should not be used in children <2 months of age or any child with suspicion of urosepsis or pyelonephritis; [b] Dosing based on trimethoprim component.

Table 4: Preferred Therapy for Unusual Pathogens

ORGANISM	DRUG OF CHOICE FOLLOWING ORGANISM IDENTIFICATION	DRUG OF CHOICE FOLLOWING SUSCEPTIBILITY RESULTS
Citrobacter sp.*, *Enterobacter sp.*, or *Acinetobacter sp.	**IV:** Cefepime **PO:** TMP/SMX	**PO TMP/SMX** *Alternatives if TMP/SMX resistant:* PO Ciprofloxacin or PO Levofloxacin
ESBL-Producing Organism	**IV Ertapenem**	**Cystitis:** PO Nitrofurantoin **Pyelonephritis:** PO TMP/SMX *Alternatives if TMP/SMX resistant:* PO Ciprofloxacin or PO Levofloxacin
Enterococcus faecalis	**Cystitis**: PO Nitrofurantoin **Pyelonephritis**: PO Amoxicillin	**Cystitis**: PO Nitrofurantoin **Pyelonephritis**: PO Amoxicillin
Vancomycin-resistant *Enterococcus faecium* (VRE)	**Cystitis**: PO Nitrofurantoin **Pyelonephritis**: PO Linezolid	**Cystitis**: PO Nitrofurantoin **Pyelonephritis:** PO Linezolid *Alternative for cystitis if nitrofurantoin resistant:* PO Linezolid

TMP/SMX = trimethoprim/sulfamethoxazole; Refer to **Table 3** for guidance on appropriate dosing of agents.

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